

Tunajenga Engineering Copilot

Complete Architecture

From Structural Questions & Site Protocols to Trusted Engineering Guidance — Powered by Self-RAG, OpenAI & Civil Standards Search

Deployed on DigitalOcean
using Docker & Compose

1. USERS

- Field Engineers**
(Ask site protocols)
- Structural Team**
(Verify blueprints)
- Project Managers**
(Check compliance)

2. FRONTEND UI

WEB UI

Tunajenga Command Center

"How to verify rebar spacing for continuous beam B-302?"

- Modern UI (HTML/CSS/JS)
- Real-time chat with citations
- Upload CAD / PDF blueprints
- Live Self-RAG execution trace
- SQLite session memory

3. BACKEND API

FASTAPI

- /chat** Engineering Copilot
- /upload** Site drawings/SOPs
- /ingest** Parse CAD & specs
- /history** Site thread memory
- /health** System diagnostics

4. SELF-RAG WORKFLOW ENGINE

LANGGRAPH

- Load Site Session (SQLite)
- Contextualize Query (LLM)
- Retrieve Vectors (Pinecone)
- Grade Site Context (LLM)
- Generate Answer from KB
- Rewrite Engineering Query
- Search Building Codes (Tavily)
- Final Answer + Audit Trace

5. DATA INGESTION PIPELINE (SINGLE SCRIPT EXECUTION: PYTHON DATA_INGESTION.PY)

Site Files

PDF Blueprints, DXF CAD, Structural SOPs, MD Guides

Processing

Chunking • Meta Extraction • Text Cleaning • Embeddings

Pinecone DB

Namespace: tunajenga-kb

6. AI STACK & TOOLS

- OpenAI LLM:** gpt-4o / gpt-5-mini
- Embeddings:** text-embedding-3-large
- Pinecone Vector DB:** tunajenga-kb
- Tavily Search:** Building codes fallback

7. PERSISTENCE LAYER

SQLite Memory: Thread sessions, LangGraph state checkpoints.

SQLite Audit Logs: User Q&A, source citations, execution traces.

8. INFRASTRUCTURE

- Host:** DigitalOcean Droplet
- Containers:** Docker Compose
- Services:** FastAPI, Nginx UI
- Proxy:** Nginx Reverse Proxy (SSL)

9. BUSINESS OUTCOMES

Faster site error resolution

Zero unverified site guidance

Unified site blueprint KB

Scalable construction tech

Monitoring & Observability: FastAPI Application Logs • Docker Container Metrics • Vector IO Error Tracking

Security & Access Control: API Key Auth • Multi-Format CAD/PDF Upload Validation • SSL Enforced